Sarathi AI Labs

Since the audience is senior IT professionals with limited Python exposure, lets learn only what helps them become productive fast.

Day 1 — Python Foundations

1. Python Introduction

- Why Python for AI / Automation
- Where Python is used today
- Running Python scripts
- Python vs other languages

2. Environment Setup

(Team will set the environment before)

- Install Python
- Install VS Code
- Install pip packages
- Terminal basics
- Run `.py` files
- Run python code using jupyter notebook (.ipynb)

3. Python Basics

- Variables
- Data types
    - String
    - Integer

    - Float
    - Boolean
- Comments
- Printing output
- Input from user

4. String Handling

- Concatenation
- f-strings
- Common string methods
- Formatting text

5. Lists

- Create list
- Access items
- Update list
- Append / remove
- Loop through lists

6. Dictionaries (Very Important)

- Create dictionary
- Key / value pairs
- Access values
- Update values
- Nested dictionaries
- Real JSON-like examples

7. Conditional Logic

- if

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- elif
- else
- Comparison operators
- Logical operators

8. Loops

- for loop
- while loop
- break / continue
- Loop through list/dictionary

9. Functions

- def
- Parameters
- Return values
- Reusable logic

Day 2 — Real World Python

10. File Handling

- Read text file
- Write text file
- Append data
- `with open()`

11. JSON Handling (Critical)

- What is JSON

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- Parse JSON
- Convert Python dict to JSON
- Read nested API response

12. Error Handling

- try / except
- Common errors
- Debugging basics

13. Modules & Packages

- import
- pip install
- requests
- anthropic

14. API Basics

- What is API
- GET request
- POST request concept
- Status codes
- Headers
- Tokens

15. Requests Library

- Install requests
- Call public API
- Read response
- Convert response JSON

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16. Claude / Anthropic API

- API key setup
- Send prompt from Python
- Receive output
- Save output

17. Productivity with AI

- Use Claude to debug code
- Ask Claude to write snippets
- Improve Python scripts

What NOT to Teach (Waste of Time)

Avoid deep dive into:

- OOP Classes
- Decorators
- Async/Await
- Generators
- Flask/Django
- Data Structures theory
- Algorithms
- Leetcode style coding

Not needed for this batch.

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Priority Ranking (Must Teach)

Highest Priority

1. Dictionaries
2. JSON
3. Functions
4. API Calls
5. Error Handling
6. Claude API

Best Delivery Order

Environment → Basics → Lists → Dict → Loops → Functions → Files → JSON → APIs → Claude

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